Successive h-indices using OpenAlex

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Abstract

I apply Schubert's successive h-index framework to OpenAlex, computing h_2 for 20,669 institutions and h_3 for 214 countries across 17.1 million researchers. The overall institution-level ranking is, in substance, a ranking of medical school size: Medicine accounts for 44% of the author pool and structurally dominates any university-wide aggregate. Field-specific decomposition is more informative: Wageningen University leads Agricultural & Biological Sciences and Environmental Science while ranking 172nd overall; Carnegie Mellon leads Computer Science from 175th; Brazil holds two of the world's six deepest dental-research benches. I normalize for group size using Egghe's Lotkaian scaling, which accounts for 91–92% of variance in both h_2 and h_3, and find that Italy and Japan reach nearly the same h_3 from a 6× difference in institutional scale. A country-level Academic Olympics reveals the United States winning 22 of 26 fields, China sweeping applied and materials science, and the United Kingdom placing in the top three in nearly every field without leading any.

Keywords: successive h-index, OpenAlex, institutional ranking, research assessment

JEL Classification: I23 , C80

MSC Classification: 62P99 , 62-07 , 62P25

1 Introduction

In 2007, András Schubert published a paper titled "Successive h-indices." (2007) It begins:

> Hirsch's excessively discussed h-index generated a long row of interpretations, applications and derivatives within a surprisingly short period. Symptomatically, Hirsch's paper in the *Proceedings of the National Academy of Sciences of the United States of America* became the second highest cited paper of the issue in which it had been published. In this note, attention is called to an apparently so far overlooked option: h-indices themselves may form the basis of a successive series of h-indices.
>
> Given, e.g., the h-indices of the researchers of an institute (which indices will be denoted by h_1 in what follows), an index, h_2, of the institute can be determined: the institute has an index h_2 if h_2 of its N researchers have an h_1-index of at least h_2 each, and the other $N - h_2$ researchers have h_1-indices lower than h_2 each. The succession can then be continued, e.g., for networks of institutions or countries or other higher levels of aggregation.
>
> Since no sufficient data for testing the concept on the researcher-institute-country hierarchy were available, another example was chosen.

Fifteen years later, OpenAlex (Priem, Piwowar, & Orr, 2022) launched a FOSS bibliographic catalogue complete with citation indexing, authors, fields, and institutions. Recent evaluations have found it to be a broadly suitable substitute for proprietary sources such as Scopus and Web of Science at the institutional level (Alperin, Portenoy, Demes, Larivière, & Haustein, 2024; Culbert et al., 2025), and I take it here as the dataset for which Schubert wished in 2007.

In the years between, successive h-indices attracted modest but sustained attention. Egghe (2008) gave the concept a theoretical footing, showing how the exponent of the underlying Lotka distribution propagates through each successive aggregation step. Two case studies followed shortly after: Ruane and Tol (2008) computed a rational (non-integer) successive h-index across economics departments in Ireland, and Arencibia-Jorge, Barrios-Almaguer, Fernández-Hernández, and Carvajal-Espino (2008) applied Schubert's exact researcher–department–institution hierarchy to a Cuban research institute using Web of Science data. None of this prior work, however, has had access to a comprehensive, open, institution-labeled snapshot of the kind OpenAlex now provides.

Schubert suggested that h_2 of institutes would be more interesting than that of journals; Section 2 describes how I compute it at scale, and Section 3 presents the results.

2 Methodology

2.1 Data source and filtering

The underlying data is OpenAlex's public authors snapshot (Priem et al., 2022), distributed as Parquet on `s3://openalex/data/parquet/authors/` (`us-east-1`, no credentials required) and partitioned into over 1,600 daily `updated_date=` directories totalling about 53 GB. Only four columns are read from each remote file (`id`, `h_index`, `affiliations`, `topics`), and an author is retained only if all of the following hold:

- `summary_stats.h_index` is non-null and greater than zero;
- at least one entry in `affiliations` has `institution.type = 'education'`;
- at least one entry in `topics` carries a non-null field classification.

Applying these filters to the full snapshot yields 17.1 million authors, spanning 20,669 institutions and OpenAlex's current list of 26 fields. One limitation carried over from the source data: an author's `h_index` is computed by OpenAlex from all works meeting its citation criteria regardless of retraction status, which OpenAlex records inconsistently (Hauschke & Nazarovets, 2025); a small, unquantified fraction of h_1 values in the dataset may therefore be inflated by since-retracted work.

2.2 Author-level assignment

Each retained author is reduced to a single row: one institution and one field. An author may list several affiliations with `institution.type = 'education'`; rather than fractionally splitting or duplicating that author across institutions, I assign them to the one with which they were most recently affiliated. Recency is read from OpenAlex's `affiliations` field, which records, for each institution an author has ever listed, the calendar years in which they published while there; I take the institution whose most recent such year is latest, breaking ties (identical latest year) by the numerically smallest OpenAlex institution identifier. I deliberately do not use the similarly-named `last_known_institutions` field for this: despite its name, it is simply the set of affiliations listed on an author's single most recent work, carrying no per-entry recency of its own, so when an author has more than one such affiliation (e.g., a joint appointment or co-affiliated coauthors misattributed to one author) there is no way to tell which is "more recent" than another.

Field assignment works the same way but is weighted. OpenAlex tags each author with a set of topics, each carrying a `count` (roughly, the number of the author's works classified under that topic) and a parent field. I sum `count` within each field and assign the author to whichever field carries the largest total weight: their modal field by publication volume, not simply their single most granular topic. The result is an `authors` table with one row per author: `(author_id, h_index, institution_id, institution_name, field, field_name)`.

2.3 The h-index and its successive generalization

Hirsch's (2005) original index, here h_1, is computed by OpenAlex per author from citation counts and taken as given. Schubert's generalization (2007) lifts the same definition up a level of aggregation: a group of N members, each already carrying an h_{k-1} index, has an index $\mathrm{h}_k = n$ if n of its members have $\mathrm{h}_{k-1} \geq n$ and the remaining $N - n$ members have $\mathrm{h}_{k-1} < n$:

Algorithm 1 Successive h-index of a group G, given a value $v(m)$ for each member $m \in G$

1: Sort G by v descending; let $r(m)$ be m's 1-indexed rank in that order
2: **return** $\max\{\, r(m) : v(m) \geq r(m),\ m \in G \,\}$

I use this rank-based procedure, implemented as a window function in SQL ranking rows within each group and taking the largest rank whose value has not yet fallen below it, for every successive index in this paper: h_2 of an institution over the h_1 of its authors, and h_3 of a country over the h_2 of its institutions.

2.4 h_2: institution and institution×field

h_2 is computed twice, over different groupings of the same author-level data. The field-specific table groups authors by (institution, field) and yields 288,300 pairs; institutions are matched on their numeric identifier only; institution and field *names* are resolved afterward via `arg_max`, so that spelling or formatting variants of the same name across snapshot partitions cannot silently split one institution into several groups. The institution-overall table ignores field entirely and applies the same procedure across all of an institution's authors regardless of discipline, yielding one h_2 per institution (20,669 rows).

2.5 h_3: country level

Institutions are mapped to countries using the `country_code` field of OpenAlex's institutions snapshot (also read directly from S3, joined on institution ID). h_3 is then computed with the same rank-based procedure, once globally per country over the h_2 values of its institutions and once per (country, field) over the field-specific h_2 values, producing one file per field.

2.6 Efficiency

Mryglod, Holovatch, and Kenna (2022) showed that group-level h-indices grow with group size even when per-member quality is held constant. They were not studying successive h indices, but a similar argument makes raw h_2 and h_3 values unsuitable for comparing institutions or countries of unequal scale.

Egghe (2008) applied a Lotkaian framework to successive h indices. Suppose the article–citation frequency function follows Lotka's law with exponent $\alpha_0 > 1$, and the author–article frequency function follows Lotka's law with exponent $\alpha_1 > 1$. Egghe showed that the distribution of h_1 values across authors is itself Lotkaian with compound exponent $\beta_1 = \alpha_0\alpha_1$. Applying the h-index formula for Lotkaian systems, the institution-level index satisfies

$$h_2 = S^{1/\beta_1} \tag{1}$$

where S is the number of authors at the institution. An analogous argument at the next level gives the distribution of h_2 values across institutions a Lotkaian exponent $\beta_2 = \alpha_0\alpha_1\alpha_2$, where α_2 is the Lotka exponent of the institution–author frequency function, so that

$$h_3 = R^{1/\beta_2} \tag{2}$$

where R is the number of institutions in the country. I define size-normalized efficiency metrics by dividing out these predicted scalings:

$$\varepsilon_2 = \frac{h_2}{S^{1/\beta_1}} \tag{3}$$

$$\varepsilon_3 = \frac{\mathrm{h}_3}{\mathrm{R}^{1/\beta_2}} \tag{4}$$

An institution with $\varepsilon_2 > 1$ outperforms the Lotkaian prediction for its headcount; one with $\varepsilon_2 < 1$ underperforms it.

The exponents β_1 and β_2 are estimated directly from the data rather than assembled from separately fit α_0, α_1, α_2. I apply discrete maximum-likelihood tail estimation following Clauset, Shalizi, and Newman (2009), with $x_{\min}$ chosen to minimize the Kolmogorov–Smirnov distance between the empirical and fitted CDFs, to the distributions of h_1 across all authors and h_2 across all institutions, reading β_1 and β_2 as their fitted Lotka exponents. I then fit α_0, α_1, and α_2 individually: α_0 from a stratified sample of per-article citation counts drawn from the OpenAlex works snapshot, α_1 from per-author work counts, and α_2 from per-institution author counts. Finally, I verify that $\alpha_0\alpha_1 \approx \beta_1$ and $\alpha_0\alpha_1\alpha_2 \approx \beta_2$ as a cross-check that the compounding-exponent structure of Egghe's model holds in the OpenAlex dataset.

2.7 Supplementary metrics

Several secondary statistics are derived from the h_2 and h_3 tables to probe what the headline numbers obscure:

- **Gini coefficient**, computed on the within-field distribution of institutional h_2, quantifies whether a field's excellence is concentrated in one institution (Gini near 1) or broadly distributed (Gini near 0).
- **Breadth score** counts, per institution, how many of the 26 fields it ranks in the global top 10, top 50, and top 100.
- **Non-biomedical ranking** recomputes institution-level h_2 after excluding authors whose modal field is Medicine, Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, or Health Professions, to test whether the overall ranking is, in substance, a ranking of medical school size.
- **Academic Olympics** converts each field's country-level h_3 ranking into an Olympic-style medal table: the top three distinct h_3 values in a field are gold/silver/bronze, ties at the same h_3 share a medal and the next tier is skipped (as in the real Olympics), and medals are tallied across all 26 fields per country.

2.8 Manual cross-validation

Because errors in OpenAlex's underlying author disambiguation or citation counts would propagate directly into every successive index, I manually spot-check a targeted, non-random sample of authors rather than a random one, chosen at the points where an error would be both most likely and most consequential: authors sitting exactly at an institution's h_2 threshold, where a ± 1 error in a single h_1 could change the institution's rank; top authors in fields where OpenAlex's topic classification is known to be thinner (e.g., Arts and Humanities); top authors at institutions with non-Anglophone naming conventions, where author-name disambiguation is more error-prone and where OpenAlex's coverage is known to diverge unevenly from other databases by country, most severely for China (Zheng, Miao, Bu, & Larivière, 2025), including institutions suspected of affiliation-swapping for grant purposes (Bhattacharjee, 2011); and the small number of authors with the highest h_1 globally, as a sanity check at the extreme of the distribution. For each candidate, I resolve the author's OpenAlex profile through the public API and compare it by hand against Web of Science, Scopus, and Google Scholar.

2.9 Software and reproducibility

All analyses were implemented in Python 3.10 (NumPy, pandas, geopandas, matplotlib, duckdb, powerlaw, scipy). The data fetching, filtering, and processing are fully reproducible from the public code repository at https://github.com/rupumped/successive-h-indices.

3 Results

3.1 Manual cross-validation

Of the 19 authors sampled under the scheme described in Subsection 2.8, Table 1 summarizes agreement between OpenAlex's `h_index` and the corresponding value I located by hand in Web of Science, Scopus, and Google Scholar. Coverage itself varies sharply: Scopus returned a value for 16 of the 19 authors, Google Scholar for 12, and Web of Science for only 11.

Table 1 Agreement between OpenAlex and each external source, over authors for which that source returned a value. Diff is OpenAlex minus the external value.

Source	n	Mean diff	Mean abs. diff	Worst case
Web of Science	11/19	+27.2	29.0	Lei Jiang (+100)
Scopus	16/19	+6.7	13.9	Donald Small (+46)
Google Scholar	12/19	−21.9	34.6	A. Jafari (−85)

Scopus tracks OpenAlex most closely, both in coverage and in magnitude of disagreement. Google Scholar is systematically higher than OpenAlex, consistent with its broader crawl of preprints, theses, and other grey literature and its comparative permissiveness toward self-citation; this ordering (Google Scholar > Scopus > OpenAlex > Web of Science) reproduces cleanly among the five global top-h_1 sanity-check authors, none of whom show a discrepancy exceeding 27% of their OpenAlex value in either direction.

Four authors, however, show disagreement well outside this range. I read this as a signature of author-disambiguation failure (multiple distinct people merged under one OpenAlex author ID) rather than of genuine cross-database disagreement:

- **Donald Small** (Johns Hopkins): $h_1^{\text{OpenAlex}} = 103$ against Scopus = 57 and Google Scholar = 59.
- **Karen S. Calhoun** (University of Michigan): $h_1^{\text{OpenAlex}} = 45$ against Scopus = 11 and Google Scholar = 13.

- **Lei Jiang** (University of Science and Technology of China): $h_1^{\text{OpenAlex}} = 215$ against Web of Science = 115, consistent with documented China-specific coverage discrepancies in OpenAlex (Zheng et al., 2025).
- **A. Jafari** (Sharif University of Technology): $h_1^{\text{OpenAlex}} = 139$, but Web of Science, Scopus, and Google Scholar report 41, 133, and 224 respectively.

All four fall in the non-Anglophone-institution or common-name strata of the sample, matching the concern raised in Subsection 2.8 that disambiguation is more error-prone for exactly these authors. Because such merges can only inflate an author's apparent h_1, and h_1 feeds directly into h_2 and h_3, a handful of merged identities at the right institution could measurably distort an institution- or country-level ranking; I flag this as an open limitation rather than one this sample can size definitively.

The two Arts and Humanities threshold authors sampled (Gillian Ramchand and G. Herbert Fowler, both sitting at Oxford's field-specific $h_2 = 30$ threshold) returned no value in any of the three external sources. This is consistent with the well-documented weaker coverage of book-based humanities scholarship in all major citation databases (Nederhof, 2006) and suggests that h-index comparisons in these fields should be read with correspondingly less confidence than in the sciences.

3.2 The overall h_2 ranking

Table 2 lists the ten institutions with the highest institution-level h_2, computed over all 20,669 institutions and 17.1 million authors described in Section 2.

Table 2 Top 10 institutions by overall h_2.

Institution	h_2	Authors
Harvard University	137	110,979
Stanford University	107	65,197
University of Cambridge	105	52,307
Cornell University	104	70,718
Johns Hopkins University	103	80,028
University of Oxford	103	57,110
University of Washington	102	71,270
University of North Carolina at Chapel Hill	101	64,478
University College London	100	77,482
University of California, San Francisco	100	50,351

This broadly tracks the US News and World Report ranking, where Harvard also places 1st, Stanford 3rd, and Cambridge 5th. As I show below, however, the overall ranking is a considerably less interesting object than it first appears.

3.3 Field-level heterogeneity

Splitting h_2 by (institution, field) reveals that the overall ranking is deceptive: several institutions that never come close to the top of Table 2 are, in fact, the single strongest research group on Earth in their field. Table 3 lists a sample of these cases.

Table 3 Selected field leaders that rank well outside the overall top 10. "Overall" columns give the institution's rank and h_2 in Table 2's full ranking; "Field h_2" is the institution's h_2 restricted to that field.

Field	Leader	Ov. rank	Ov. h_2	Field h_2
Agricultural & Biological Sciences	Wageningen	172	71	56
Environmental Science	Wageningen	172	71	56
Pharmacology, Toxicology & Pharmaceutics	Shenyang Pharmaceutical University	535	53	24
Computer Science	Carnegie Mellon University	175	70	56
Materials Science	USTC	39	89	59
Engineering	USTC	39	89	62
Energy	USTC	39	89	52
Chemical Engineering	Univ. of Chinese Academy of Sciences	97	80	23
Dentistry	Universidade de São Paulo	235	66	34

University of Antwerp is the counterexample that proves the rule holds both ways: it is #14 overall ($h_2 = 98$) *and* #1 in Physics and Astronomy ($h_2 = 96$), a broad powerhouse that also happens to lead a specific, non-biomedical field outright. Universidade Estadual de Campinas (UNICAMP) places 6th in Dentistry behind São Paulo, giving Brazil two of the world's six deepest dental-research benches despite the country appearing nowhere near the top of any general-purpose ranking (Vaccaro, Sánchez-Núñez, & Witt-Rodríguez, 2022). China's cluster is not one specialty but four related applied-science fields: University of Science and Technology of China (USTC) leads Materials Science, Engineering, and Energy outright, while its sister institution, the University of Chinese Academy of Sciences, leads Chemical Engineering.

Not every field supports this kind of comparison equally well. Decision Sciences is the thinnest field in the dataset: Stanford's global #1 is $h_2 = 22$, built on only 285 Stanford authors against 139,954 authors field-wide, versus Medicine's $h_2 = 119$ built on 7.58 million authors field-wide. h_2 is not meaningfully comparable across fields of such different size, only within a field, and Decision Sciences in particular may be too sparse an OpenAlex topic category for its ranking to carry much signal.

3.4 Concentration within fields

Table 4 reports, for every field, the Gini coefficient of institutional h_2 defined in Section 2.7, together with the field's institution count, leading (maximum) h_2, and mean h_2.

Table 4 Gini coefficient of institutional h_2 within each field, sorted from most to least concentrated.

Field	Institutions	Max h_2	Mean h_2	Gini
Physics and Astronomy	12,757	96	6.10	0.605
Biochemistry, Genetics & Molecular Biology	15,395	104	6.62	0.577
Neuroscience	11,656	64	4.41	0.574
Immunology and Microbiology	9,842	67	3.97	0.561
Materials Science	13,085	59	6.37	0.560
Earth and Planetary Sciences	12,487	60	4.29	0.557
Chemistry	12,107	41	5.02	0.541
Environmental Science	16,603	56	5.55	0.541
Engineering	17,734	62	6.91	0.533
Medicine	19,137	119	8.57	0.528
Agricultural and Biological Sciences	16,401	56	5.60	0.522
Energy	8,957	52	3.39	0.515
Psychology	15,745	58	3.64	0.513
Computer Science	16,747	56	5.08	0.512
Dentistry	8,291	34	3.01	0.500
Mathematics	11,049	36	3.43	0.491
Economics, Econometrics and Finance	13,940	55	3.30	0.482
Business, Management and Accounting	14,620	32	3.59	0.481
Health Professions	14,863	41	2.88	0.465
Social Sciences	19,258	48	4.56	0.461
Pharmacology, Toxicology and Pharmaceutics	7,164	24	2.63	0.459
Chemical Engineering	5,880	23	2.49	0.445
Arts and Humanities	15,436	30	2.77	0.443
Nursing	8,951	31	2.37	0.433
Decision Sciences	10,385	22	2.37	0.423
Veterinary	4,727	28	2.03	0.420

Physics and Astronomy is the most unequal field: University of Antwerp's $h_2 = 96$ is 15.7 times the field mean of 6.1. Veterinary and Decision Sciences are the most evenly distributed (Gini $\approx$ 0.42). Medicine sits in the middle of the table (0.528) despite Harvard's large absolute lead, because Medicine, unlike Physics and Astronomy, has thousands of institutions with meaningful h_2 rather than a single dominant outlier.

3.5 The overall ranking as an implicit biomedical ranking

Medicine alone accounts for 7.58 million of the dataset's 17.1 million authors (44%), and, together with Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, and Health Professions, the five core biomedical fields dwarf every other field's author pool. Table 5 recomputes institution-level h_2 after excluding authors in these five fields entirely to test whether the overall ranking in Table 2 is, in substance, a ranking of medical school size.

Table 5 Top 10 institutions by non-biomedical h_2, alongside their position in the unrestricted overall ranking. Δrank is overall rank minus non-biomedical rank; positive means the institution moves up once biomedical fields are excluded.

Institution	Rank	h_2	Overall rank	Overall h_2	Δrank
Harvard University	1	100	1	137	0
University of Antwerp	2	97	14	98	+12
California Institute of Technology	3	94	15	97	+12
Stanford University	4	92	2	107	−2
Université Paris-Saclay	5	91	22	95	+17
Massachusetts Institute of Technology	6	89	25	94	+19
University of Geneva	7	88	29	93	+22
Univ. of Science and Technology of China	8	88	39	89	+31
ETH Zurich	9	87	36	89	+27
University of Cambridge	10	87	3	105	−7

Harvard remains #1 even with its biomedical authors removed, confirming the claim in Subsection 3.3 that it is genuinely excellent rather than merely large. But the rest of the table reshuffles substantially: Antwerp, Caltech, Paris-Saclay, MIT, Geneva, USTC, and ETH Zurich all gain 12 to 31 places once Medicine and its four sibling fields are set aside, while Stanford and Cambridge, both anchored by large medical schools, fall by 2 and 7 places respectively. The University of Chinese Academy of Sciences shows the same pattern one tier down: 97th overall but 23rd non-biomedical. Because medical fields have by far the largest, most-cited author pools, they structurally dominate any university-wide aggregate; an institution elite in non-medical fields, such as USTC or Carnegie Mellon, can never reach the top of the overall list no matter how good it is, simply because it lacks a large medical school. The field-level and non-biomedical breakdowns matter more than the unrestricted university-level number.

3.6 Breadth versus depth

Table 3 showed institutions that lead a single field while sitting far outside the overall top 100. Table 6 shows the opposite pattern: institutions whose strength is broad rather than deep, measured by how many of the 26 fields they place in the global top 50.

UCLA has the greatest breadth by this measure, placing in the global top 50 in 20 of 26 fields despite leading only 8 outright, more consistently broad than even Harvard's

Table 6 The eight institutions with the most top-50 field placements out of 26 fields.

Institution	Top 50	Top 10	Overall rank	Overall h_2
University of California, Los Angeles	20	8	13	99
Harvard University	19	13	1	137
University of Oxford	19	8	6	103
University of Toronto	17	7	11	100
Cornell University	17	6	4	104
University of Michigan	17	6	28	94
Univ. of North Carolina at Chapel Hill	17	5	8	101
University of Cambridge	17	4	3	105

19. Harvard still leads more fields outright (13 of 26) than any other institution, but four of the eight broadest institutions by this top-50 count, Toronto, Michigan, UNC Chapel Hill, and Cambridge, place fewer than half their 17 top-50 fields inside the top 10, meaning most of their breadth comes from being solidly good across many fields rather than dominant in any one. At the other extreme, several institutions post a high overall h_2 without ever leading a single field: UC San Diego (rank 17, $h_2 = 97$), the University of Melbourne (rank 23, $h_2 = 95$), and Boston University (rank 25, $h_2 = 94$) all have zero top-10 field placements. These institutions achieve their overall strength through scale and consistency across many merely-good departments, not through fielding the single best group in any one discipline.

3.7 Institution efficiency

Model validation

Figure 1 plots institution-level h_2 against the Egghe-predicted value S^{1/β_1} for all institutions with at least 10 OpenAlex authors. A no-intercept regression, appropriate because $N = 0$ authors forces $h_2 = 0$, gives slope $c = 1.945$ and uncentered $R^2 = 0.911$: S^{1/β_1} alone explains 91% of the variance in h_2, confirming that Egghe's power-law functional form describes the data. The efficiency metric $\varepsilon_2 = h_2/S^{1/\beta_1}$ captures each institution's deviation from this overall trend.

The cross-check on Egghe's compounding structure fails sharply, however. Egghe's derivation requires that the Lotka tail exponent of the h_1 distribution equals the product $\alpha_0\alpha_1$; these separate fits give $\alpha_0 = 2.742$ and $\alpha_1 = 2.996$, whose product is 8.21, against the directly-measured $\beta_1 = 2.963$: a discrepancy of 177%. The analogous prediction at the next level ($\beta_2 = \beta_1\alpha_2$) is similarly violated: $2.963 \times 1.994 = 5.909$ against the directly-measured $\beta_2 = 2.998$ (97% discrepancy).

Efficiency rankings

The efficiency metric $\varepsilon_2 = h_2/S^{1/\beta_1}$ addresses Mryglod et al.'s (2022) concern directly: because S^{1/β_1} accounts for 91% of the variance in h_2, dividing it out isolates the residual variation that is not explained by headcount alone.

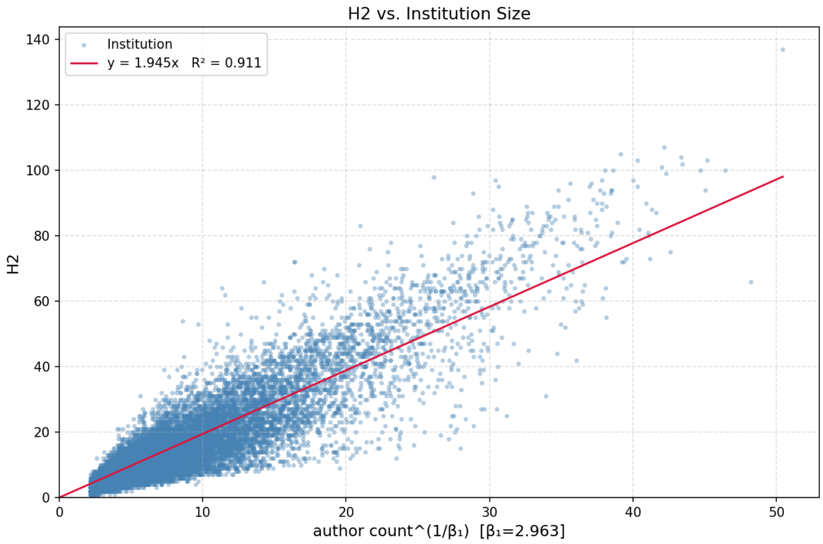

Fig. 1 Institution-level h_2 versus S^{1/β_1} (Egghe's predicted scaling, $\beta_1 = 2.963$). No-intercept regression: $h_2 = 1.945\, S^{1/\beta_1}$, $R^2 = 0.911$ (uncentered).

Table 7 lists the top 10 institutions by ε_2 among the 12,026 with at least 100 authors. The top of the ranking is dominated by small and highly specialized institutions. Focused research centers and specialized colleges recur throughout the top 20.

Table 7 Top 10 institutions by efficiency $\varepsilon_2 = h_2/S^{1/\beta_1}$ ($\beta_1 = 2.963$) among those with at least 100 authors ($n = 12{,}026$).

Institution	ε_2	h_2	Authors	h_2 rank
Bellevue University	6.27	54	590	497
Augustana University	5.65	64	1,332	264
Center for Behavioral Brain Sciences	5.50	31	168	1,908
Barcelona Inst. of Science and Technology	5.46	53	839	520
Imperial Valley College	5.36	62	1,412	319
Saint Camillus International Univ. of Health Sci.	5.33	39	364	1,228
City College of San Francisco	4.95	43	605	934
Frederick Community College	4.83	37	416	1,331
Académie Nationale de Médecine	4.75	39	512	1,192
Natl. Interuniversity Consortium of Materials Sci.	4.70	40	571	1,150

Table 8 restricts to the 1,317 institutions with at least 5,000 authors. The leading institutions are now a mix of specialized research institutes like Rockefeller University ($\varepsilon_2 = 3.79$, biomedical), the Institute of Cancer Research ($\varepsilon_2 = 3.59$), and the London School of Hygiene & Tropical Medicine ($\varepsilon_2 = 3.47$) and comprehensive universities like the University of Antwerp ($\varepsilon_2 = 3.75$, $h_2 = 98$, efficiency rank 110). The University of

California System, treated here as a single institution under its OpenAlex identifier, is the only entry in the top 100 on both metrics simultaneously (h_2 rank 73, efficiency rank 66). Among the comprehensive elite research universities that dominate the raw h_2 table, Caltech (efficiency rank 363), the University of Geneva (334), and Université Paris-Saclay (450) are the most efficient, each achieving $h_2 \geq 93$ with under 25,000 authors.

Table 8 Top 10 institutions by efficiency $\varepsilon_2 = h_2/S^{1/\beta_1}$ ($\beta_1 = 2.963$) among those with at least 5,000 authors ($n = 1{,}317$).

Institution	ε_2	h_2	Authors	h_2 rank
University of California System	3.96	83	8,242	73
Rockefeller University	3.79	70	5,637	179
University of Antwerp	3.75	98	15,759	14
Institute of Cancer Research	3.59	66	5,570	230
University of San Diego	3.59	64	5,097	286
California University of Pennsylvania	3.52	76	9,011	121
University of Wuppertal	3.51	68	6,515	209
London School of Hygiene & Tropical Medicine	3.47	65	5,903	249
Royal Holloway, University of London	3.42	61	5,102	339
Vita-Salute San Raffaele University	3.40	71	8,171	171

3.8 h_3: country level

Aggregating institutional h_2 up to the country level via the same rank-based procedure yields h_3. Table 9 lists the ten highest-h_3 countries, and Figure 2 maps h_3 for all 214 countries with at least one qualifying institution.

Table 9 Top 10 countries by h_3.

Country	h_3	Institutions
United States	70	4,242
China	55	1,078
Germany	49	505
United Kingdom	47	548
Japan	46	1,358
Italy	45	229
France	41	394
Spain	39	156
South Korea	37	358
Australia	36	120

The United States leads by a wide margin, but Italy's 45 on only 229 institutions and Japan's 46 on 1,358 are a first hint that raw h_3 rewards national research-system size as well as depth, a point to which I return in Subsection 3.10.

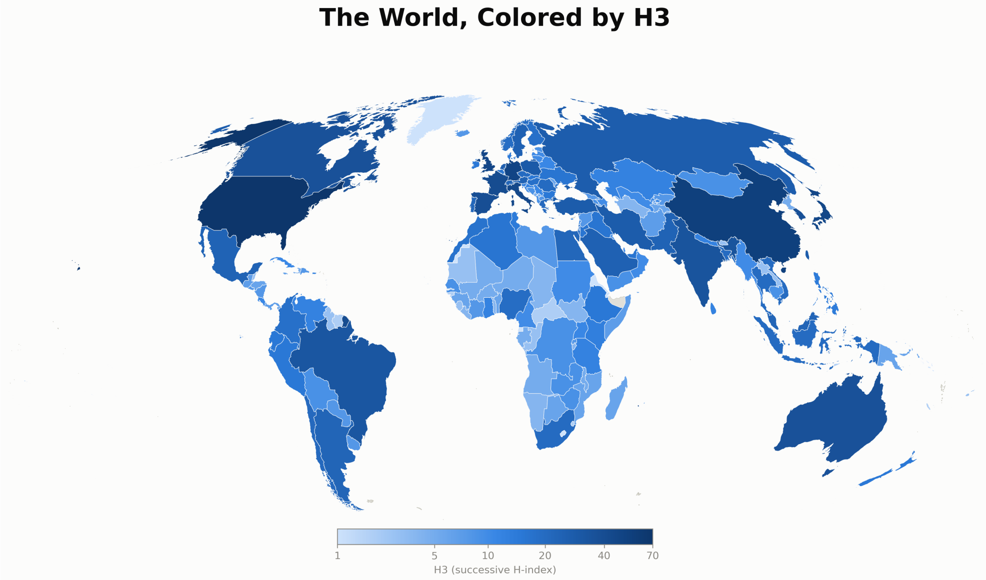

Fig. 2 h_3 by country. Countries with no qualifying institution are shown in grey.

3.9 Academic Olympics

Section 2.7 defined an Olympic-style medal table by awarding gold/silver/bronze to the top three distinct h_3 tiers in each of the 26 fields, with ties sharing a medal. Figure 3 shows the resulting medal table.

The United States wins 22 of 26 golds, missing only the four fields where China wins outright (Chemical Engineering, Energy, Engineering, Materials Science) and tying China for gold in Chemistry, where both countries reach $h_3 = 28$ (the US on 2,177 institutions, China on 970). China's five golds form a clean sweep of applied and materials science, with zero medals in arts, humanities, social sciences, or business, an unambiguously present-tense result rather than a "catching up" story. Great Britain is the clearest case of depth without a peak: 10 silvers and 9 bronzes, but zero golds, consistent with world-class institutions that are nonetheless outgunned by US depth in every single field.

3.10 Country efficiency

Figure 4 repeats the size-scaling analysis at the country level, plotting h_3 against R^{1/β_2} for each country. The no-intercept fit gives slope $c = 4.064$ and uncentered $R^2 = 0.922$: institution count alone explains 92% of the variance in h_3 across countries.

Table 10 ranks the 151 countries with at least 5 qualifying institutions by efficiency $\varepsilon_3 = h_3/R^{1/\beta_2}$. The Italy–Japan contrast foreshadowed in Subsection 3.8 is now quantified: Italy ($\varepsilon_3 = 7.35$, rank 1) achieves $h_3 = 45$ from only 229 institutions, while Japan ($\varepsilon_3 = 4.15$, rank 44) reaches $h_3 = 46$ from 1,358, virtually the same headline

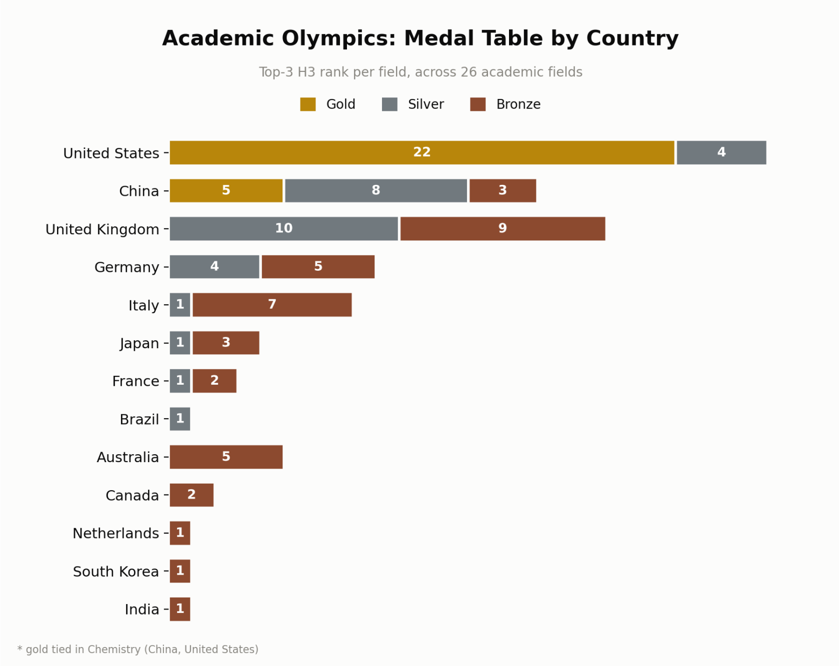

Fig. 3 Academic Olympics medal table: gold/silver/bronze by top-3 h_3 rank per field, across 26 fields.

number but at 6× the institutional scale. Australia (rank 2, $\varepsilon_3 = 7.29$, 120 institutions) and Spain (rank 3, $\varepsilon_3 = 7.24$, 156 institutions) round out a top tier of countries with compact but high-quality research systems. Germany (rank 6), the United Kingdom (rank 7), and France (rank 8) are the only countries that place in the top 10 on both raw h_3 and efficiency, confirming that their research depth is genuine rather than an artefact of institutional scale. Saudi Arabia is a notable outlier at 4th by efficiency (h_3=25 on just 52 institutions) despite ranking only 20th by raw h_3; as flagged in Subsection 2.8, this is plausibly an artifact of high-h-index researchers listing Saudi institutions as secondary affiliations in exchange for funding (Bhattacharjee, 2011) rather than a genuine concentration of resident research talent.

The United States falls to efficiency rank 34 ($\varepsilon_3 = 4.32$) despite its raw lead ($h_3 = 70$, rank 1). With 4,242 qualifying institutions, nearly 8× more than China and 18× more than Italy, even a large absolute h_3 advantage is absorbed by the size denominator. China (rank 10, $\varepsilon_3 = 5.36$, 1,078 institutions) is more compact and appears above the US on this metric.

4 Conclusion

Schubert closed his 2007 note wondering whether h_2 of institutes would prove more interesting than h_2 of journals, absent the researcher–institute–country data needed to check. Nineteen years and 17.1 million OpenAlex authors later, the answer is yes, but not in the way a single top-10 table would suggest.

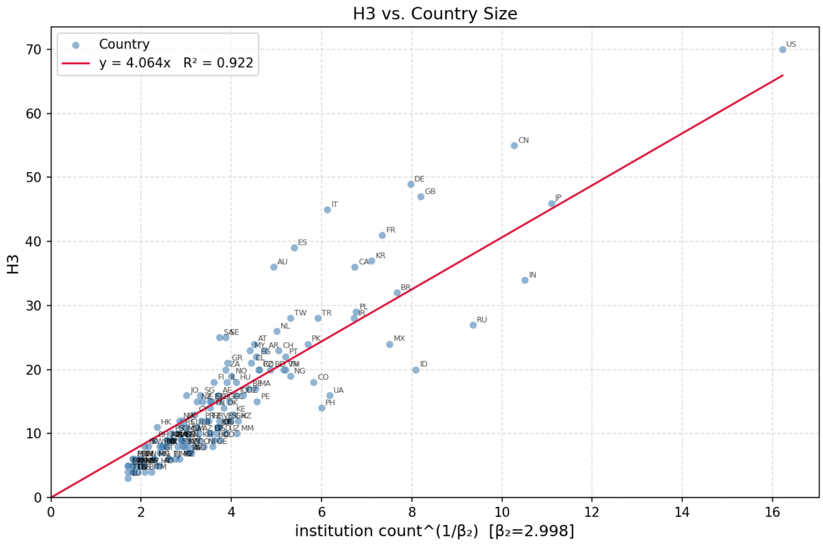

Fig. 4 Country-level h_3 versus R^{1/β_2} (Egghe's predicted scaling, $\beta_2 = 2.998$). No-intercept regression: $h_3 = 4.064\,R^{1/\beta_2}$, $R^2 = 0.922$ (uncentered).

Table 10 Top 10 countries by efficiency $\varepsilon_3 = h_3/R^{1/\beta_2}$ ($\beta_2 = 2.998$) among those with at least 5 institutions ($n = 151$).

Country	ε_3	h_3	Institutions	h_3 rank
Italy	7.35	45	229	6
Australia	7.29	36	120	10
Spain	7.24	39	156	8
Saudi Arabia	6.69	25	52	20
Sweden	6.45	25	58	21
Germany	6.14	49	505	3
United Kingdom	5.74	47	548	4
France	5.59	41	394	7
Greece	5.36	21	60	31
China	5.36	55	1,078	2

The overall h_2 ranking is the least interesting object in this paper. Because Medicine alone accounts for 44% of the author pool, the unrestricted university-wide aggregate is, in substance, a ranking of medical school size. The field decomposition reveals a completely different map of global research strength: Wageningen leads Agricultural & Biological Sciences and Environmental Science globally while ranking 172nd overall; Carnegie Mellon leads Computer Science from 175th; the University of Science and Technology of China sweeps Materials Science, Engineering, and Energy from 39th; and Shenyang Pharmaceutical University leads Pharmacology from 535th. Brazil offers perhaps the sharpest illustration of what the overall ranking hides: the

Universidade de São Paulo and the Universidade Estadual de Campinas hold two of the world's six deepest dental-research benches despite neither country nor either institution appearing anywhere near the top of any general-purpose ranking. The field-level h_2 is the unit that should be the default for comparison, not the aggregate.

The Academic Olympics is the most readable summary of the country-level results. The United States wins 22 of 26 golds. China's five represent a focused push in applied and materials science; there are no Chinese medals in arts, humanities, social sciences, or business. The United Kingdom places in the top three in nearly every field (10 silvers, 9 bronzes) and tops none of them. Italy and Japan both reach $h_3 \approx 45$–46 while differing 6× in institutional scale (229 versus 1,358 institutions); the raw ranking puts them adjacent, and the efficiency metric pulls them apart.

On the methodological side, Egghe's power-law functional form is empirically well-supported: institution count alone explains 91–92% of the variance in h_2 and h_3 respectively. However, the theoretical compounding prediction $\alpha_0\alpha_1 \approx \beta_1$ fails by 177%. The model is empirically useful; the underlying theory needs a different reading.

Several limitations constrain interpretation. Author disambiguation is the most consequential open problem: the four outlier cases identified in cross-validation (notably Lei Jiang, $h_1^{\text{OpenAlex}} = 215$ against Web of Science = 115) confirm that merger errors cluster in non-Anglophone and common-name strata, exactly the authors whose inflated h_1 values would propagate most damagingly into institutional and country-level rankings. Arts and Humanities are systematically under-covered in all major citation databases: both threshold authors sampled at Oxford returned no value in Web of Science, Scopus, or Google Scholar, and h-index comparisons in book-based fields should be read with correspondingly less confidence. OpenAlex's inconsistent retraction flagging leaves a small, unquantified fraction of h_1 values potentially inflated. Saudi Arabia's efficiency rank (4th, $\varepsilon_3 = 6.69$) is plausibly an artifact of affiliation-listing for funding rather than resident research presence (Bhattacharjee, 2011), a data-quality distortion visible in the output but not removable without ground-truth affiliation data. Finally, h_2 is not comparable across fields of different size: Decision Sciences (max $h_2 = 22$) cannot be placed on the same scale as Medicine (max $h_2 = 119$), and the results should be read within fields only.

The most impactful near-term improvement to this framework would require a data source that does not yet exist: departmental affiliation in OpenAlex. If OpenAlex were to record the department within an institution alongside the institution itself, the hierarchy would extend naturally to researcher → department → institution → country, with successive indices h_1 through h_4. Departments are field-specialized by construction, so the step of inferring each author's modal field from topic weights may become unnecessary at the departmental level. I flag this as the single most valuable addition OpenAlex could make for scientometric work of this kind.

Statements and Declarations

The author did not receive support from any organization for the submitted work. The author has no relevant financial or non-financial interests to disclose. This study did not involve human participants or animals, and thus no ethical approval or informed

consent was required. The data that support the findings of this study are publicly available from the OpenAlex public snapshot. The analysis code is available at the GitHub repository provided in the manuscript.

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